

DESIGN OF FLUIDISED BED COMBUSTION BOILERS FOR MULTIFUELS

The [original paper](#) contains 6 sections, with 10 passages identified by our machine learning algorithms as central to this paper.

Paper Summary

SUMMARY PASSAGE 1

Fuel Heat Duty

The fuels that are specified for co firing are usually Bagasse pith, Rice husk, Biogas, Sawdust, Wood chips, Coffee husk. Co firing has an advantage that the fuel quantity as received is fired directly with the support of main supporting fuel. Many times 60 % of heat input may be substituted by the auxiliary fuel.

SUMMARY PASSAGE 2

Fuel Type Fibrous / Non-Fibrous / Liquid / Gaseous

In the case of multi fuels, it is possible to have metering of all fuels. Or only the main fuel can be metered and the other fuels may be designed for direct firing. See figure 1 & 2 showing the fuel feed arrangements.

SUMMARY PASSAGE 3

Fuel Feeder Sizing

Higher the bulk density of the fuel, the greater the volume of the feeder. A fuel feeder sized for coal will be too small for rice husk. Metering coal with a feeder designed for rice husk would be difficult.

SUMMARY PASSAGE 4

Fuel Storage Bunker

For Bagasse & fibrous materials, flight chain conveyor or belt conveyors are selected to convey the fuel from the yard to boiler. In this applications a horizontal run is planned to enable feeding the conveyor anywhere along the length. When it comes to regulating the fuel feed to the boiler, the return Bagasse conveyor is planned so that surplus fuel can be returned to yard.

SUMMARY PASSAGE 5

Hours Of Operation Of Fuel Handling System

The fuel handling system will have to operate for all three shifts in the case of low bulk density / fibrous fuels.

SUMMARY PASSAGE 6

Ash Melting Point

FBC furnaces are designed for bed temperatures of 800 to 900 deg C in the case of the underbed feeding system. In the case of the over bed feeding arrangement, the over bed temperature is likely to exceed 1000 deg C. For fuels which have low ash melting points, it is desirable that Combustion temperatures are kept lower. For fuels like coffee husk, 100 % firing has not been practical for its ash melting temperature is as low as 800 deg C. In the case of Cofiring with high ash fuels such as coal, rice husk, the slagging is greatly reduced.

SUMMARY PASSAGE 7

Sulfur Content In Fuel

Table 8 suggests the % ash collection rate and sizing philosophy for ash feeders for Bi drum type boiler. Bed ash removal is dependent on the size of the fuel particles and the fuel itself. For fuels like coal, bed ash cooling system shall be sized for 40 %.

SUMMARY PASSAGE 8

â€¢ Ash Handling System & Ash Silo

The bed ash is negligible in the case of the Rice husk and other agro wastes. Hence for fossil fuels such as coal / lignite only the bed ash cooling system & bed ash conveying systems are planned. For fly ash, lean phase system and dense phase system are preferred as the dust nuisance is less.

SUMMARY PASSAGE 9

Furnace Heat Release Rate

Refractory lining will be required to improve the Furnace outlet temperature in order to obtain desired convection superheater outlet steam temperature. Traveling grate furnace is not the right solution for coal / Bagasse or rice husk/ Bagasse firing. Because it is not possible to obtain the steam temperature as the combustion temperature is different for the different fuels and the furnace heat absorption pattern is different between high & low moisture fuels.

SUMMARY PASSAGE 10

Bed Coils Surfacing & Excess Air.

In under bed FBC system the bed coil surfacing is done based on the fuel to be fired. The fuel chemical composition and the volatile matter are the deciding factors. Table 11 shows a heat balance of the bed for selected fuels.